

COMODORO RIVADAVIA, Argentina

WMO: 878600

Lat: 45.7922S

Lon: 67.4619W

Elev: 46

StdP: 100.77

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.2	0.0	-13.1	1.2	14.7	-10.9	1.5	12.9	18.0	11.3	16.2	11.2	3.7	270	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	31.0	16.1	28.9	15.4	27.1	14.6	17.8	27.1	16.7	25.5	15.8	24.1	8.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.8	10.6	19.7	13.3	9.6	18.7	12.1	8.8	18.0	50.5	27.0	47.0	25.7	44.3	24.2	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.4	13.0	-3.8	35.4	1.6	2.1	-4.9	36.9	-5.8	38.1	-6.7	39.3	-7.8	40.8		
(5)			-4.8	19.9	1.3	1.5	-5.8	21.0	-6.6	21.9	-7.3	22.7	-8.3	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	19.5	18.6	16.6	13.6	10.3	7.6	7.2	8.5	10.3	13.1	15.7	18.3		
	DBStd	5.53	3.35	3.77	3.59	3.51	3.51	3.53	3.50	3.28	3.52	3.45	3.45	3.43		
	HDD10.0	354	0	0	1	7	39	88	100	69	38	12	1	0		
	HDD18.3	2054	26	38	76	146	250	322	346	306	241	167	93	44		
	CDD10.0	1542	295	242	207	117	48	16	12	21	48	106	173	257		
	CDD18.3	199	62	47	23	6	1	0	0	0	1	3	14	42		
	CDH23.3	1732	547	432	192	34	1	0	0	0	5	24	132	364		
	CDH26.7	497	170	147	47	4	0	0	0	0	0	2	28	97		
(14) Wind	WSAvg	6.0	6.9	6.0	5.6	5.8	5.2	5.6	5.5	5.6	5.6	6.1	6.8	7.0		
(15) Precipitation	PrecAvg	237	14	16	19	28	33	32	23	22	16	13	11	12		
	PrecMax	442	74	90	96	155	150	170	131	84	92	58	53	112		
	PrecMin	123	0	0	0	0	0	2	0	1	0	0	0	0		
	PrecStd	76	17	19	21	33	33	33	24	22	16	12	13	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	34.1	30.8	27.0	21.8	18.2	17.9	19.5	23.8	26.1	30.0	32.1		
		MCWB	17.2	18.3	16.1	14.2	12.3	9.6	9.8	10.3	11.5	13.2	15.3	16.2		
	2%	DB	30.8	30.7	27.8	23.9	19.3	15.9	15.7	17.1	20.1	23.1	26.7	29.1		
		MCWB	16.2	16.4	15.1	13.6	11.2	9.0	8.7	9.0	10.2	11.8	13.8	15.3		
	5%	DB	28.2	28.1	25.2	21.4	17.9	14.2	14.0	15.2	18.1	21.1	24.5	27.0		
		MCWB	15.0	15.8	14.1	12.3	10.7	8.1	7.8	8.0	9.1	10.7	12.9	14.4		
	10%	DB	26.2	25.9	23.2	19.8	16.1	12.9	12.5	13.9	16.4	19.2	22.6	25.1		
		MCWB	14.1	14.6	13.3	11.4	9.6	7.3	6.9	7.4	8.2	9.9	11.9	13.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	19.6	17.7	15.8	13.6	10.8	10.7	11.4	12.5	14.6	16.4	18.1		
		MCDB	29.4	30.2	27.1	23.7	19.7	16.4	16.2	17.9	21.1	23.0	25.9	29.1		
	2%	WB	17.7	17.9	16.2	14.4	12.3	9.6	9.4	9.9	10.9	12.8	14.8	16.4		
		MCDB	26.4	27.1	23.9	21.4	17.9	14.5	14.5	15.8	18.2	20.8	24.4	26.4		
	5%	WB	16.4	16.7	15.3	13.2	11.2	8.7	8.3	8.9	10.0	11.7	13.6	15.2		
		MCDB	25.0	25.2	22.6	19.4	16.5	13.4	12.9	14.3	16.6	19.2	22.7	24.8		
	10%	WB	15.3	15.6	14.3	12.2	10.2	7.7	7.3	7.9	9.0	10.7	12.5	14.3		
		MCDB	24.2	23.4	21.5	18.2	14.8	11.9	11.7	12.5	15.0	17.8	20.9	23.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	10.8	10.3	9.3	7.9	7.0	7.3	8.1	9.1	10.2	11.0	11.2		
		MCDBR	14.3	14.8	13.5	11.7	9.7	7.9	8.5	9.6	11.5	12.6	13.9	14.1		
		MCWBR	5.9	6.1	6.1	6.0	5.6	4.9	5.0	5.3	5.8	5.9	6.4	6.1		
		MCDBR	13.0	13.1	12.0	10.7	9.0	7.6	8.1	9.0	10.7	11.8	13.2	13.2		
	5% WB	MCWBR	6.3	6.1	6.2	6.4	5.8	5.2	5.2	5.5	6.0	6.3	6.8	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.339	0.317	0.309	0.293	0.282	0.282	0.297	0.314	0.318	0.322	0.335		
	taud		2.537	2.525	2.583	2.595	2.594	2.579	2.592	2.579	2.527	2.528	2.536	2.512		
	Ebn at Noon		986	950	925	855	788	756	794	854	912	963	993	992		
	Edh at Noon		105	101	85	71	58	52	56	70	88	99	105	110		
(44) All-Sky Solar Radiation	RadAvg		7.50	6.24	4.54	2.96	1.73	1.29	1.50	2.35	3.78	5.51	6.95	7.66		
	RadStd		0.25	0.34	0.21	0.15	0.11	0.08	0.09	0.19	0.22	0.24	0.21	0.38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.35	N/A	+2.49	+0.48	+0.92	+1.13	N/A	-105	+72	+24
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page